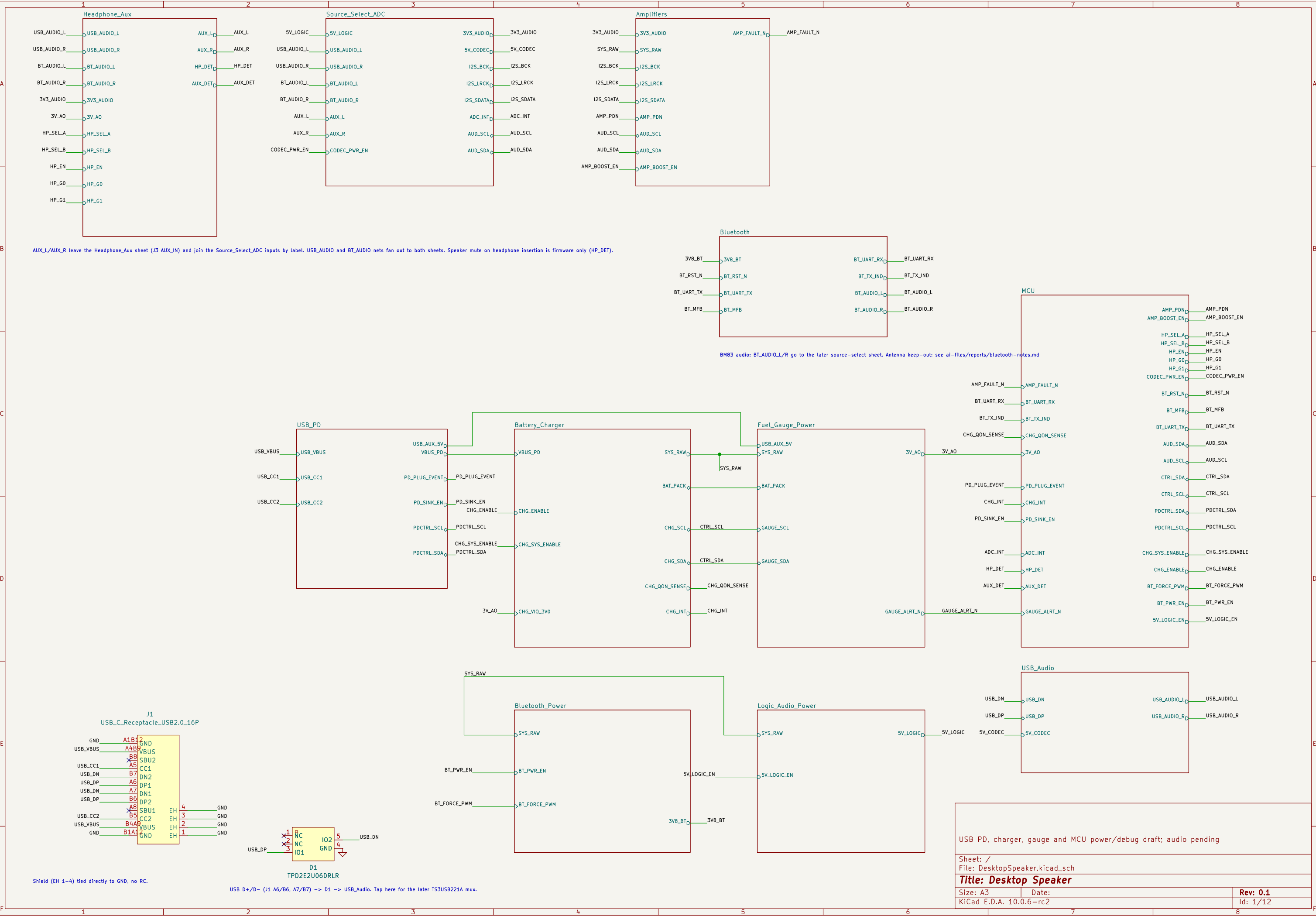
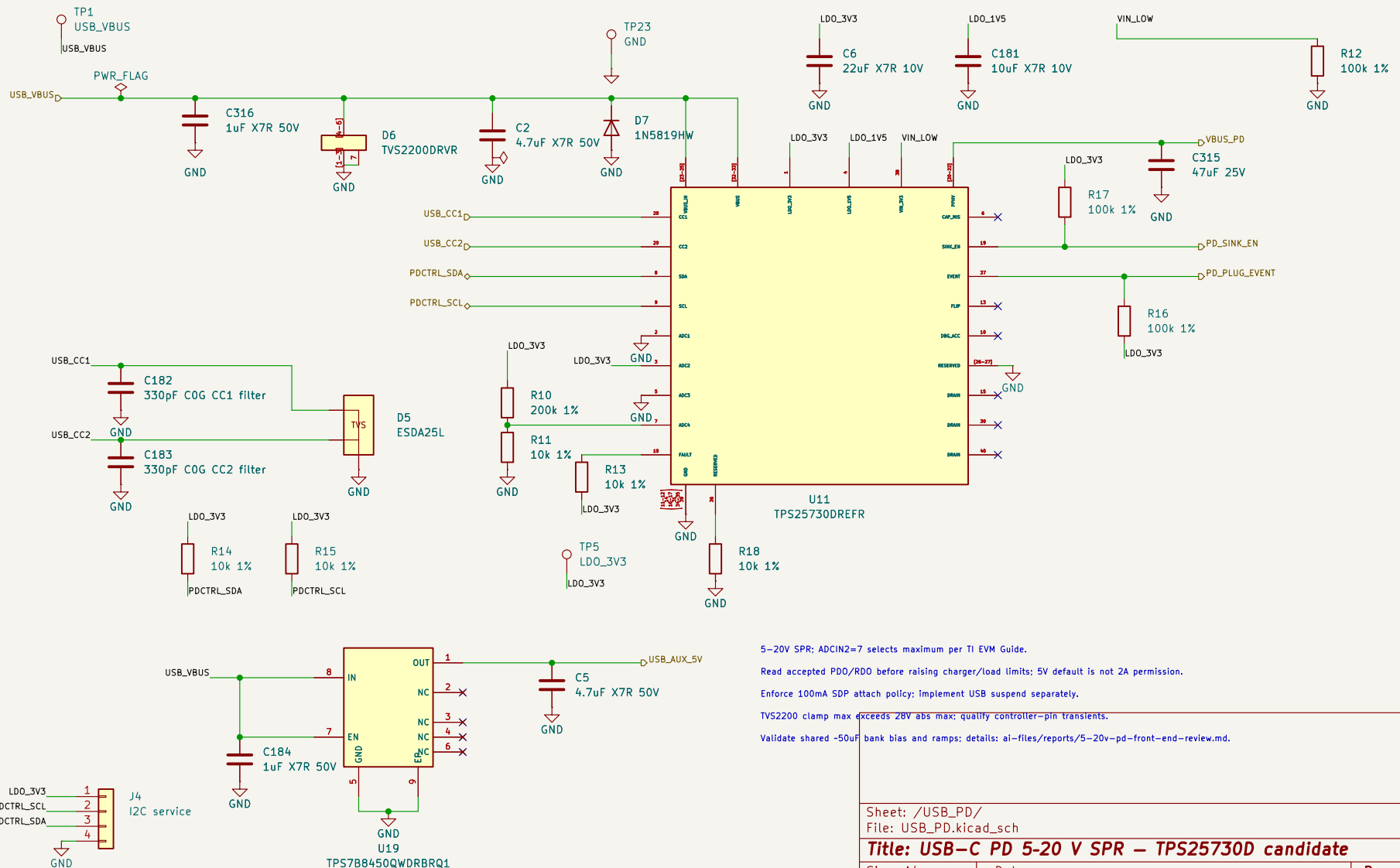






5–20V SPR; ADCIN2=7 selects maximum per TI EVM Guide.

Read accepted PDO/RDO before raising charger/load limits; 5V default is not 2A permission.

Enforce 100mA SDP attach policy; implement USB suspend separately.

TVS2200 clamp max exceeds 28V abs max; qualify controller–pin transients.

Validate shared –50uF bank bias and ramps; details: ai-files/reports/5–20v–pd–front–end–review.md.

Sheet: /USB\_PD/

File: USB\_PD.kicad\_sch

**Title: USB–C PD 5-20 V SPR – TPS25730D candidate**

Size: A4

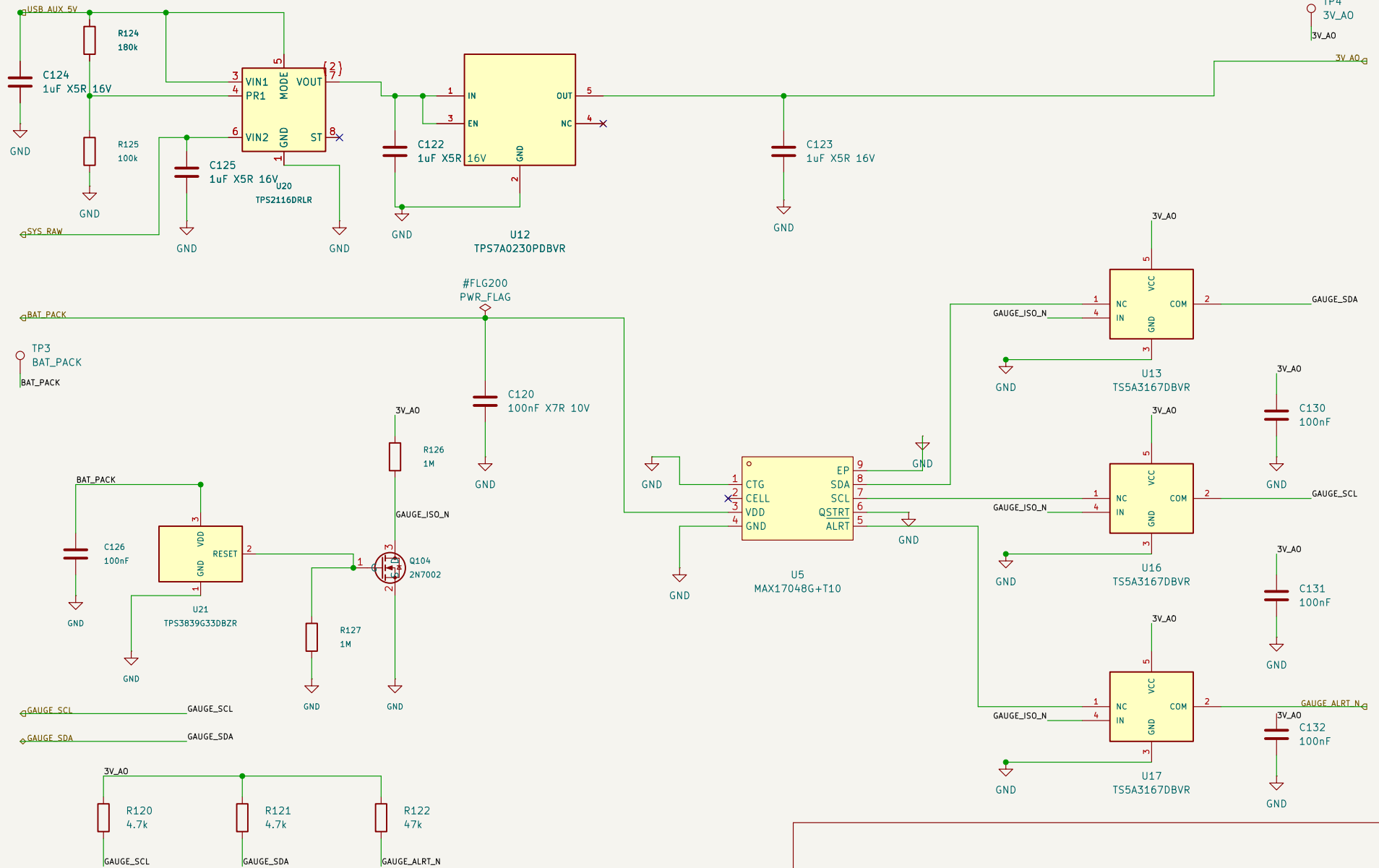
Date:

KiCad E.D.A. 10.0.6–rc2

Rev: prototype review

Id: 2/12





USB-priority logic supply; PR1 nominal 2.8 V.  
Switchover/BOR and effective capacitance need qualification.

CELL pad 2 unused on MAX17048; QSTRT tied low

Gauge bus; AO-powered switches; default OFF below valid battery.  
U21 3.08 V supervisor enables Q104 after reset recovery.

Sheet: /FuelGauge\_Power/  
File: FuelGauge\_Power.kicad\_sch

# **Title: Fuel gauge and low-current power**

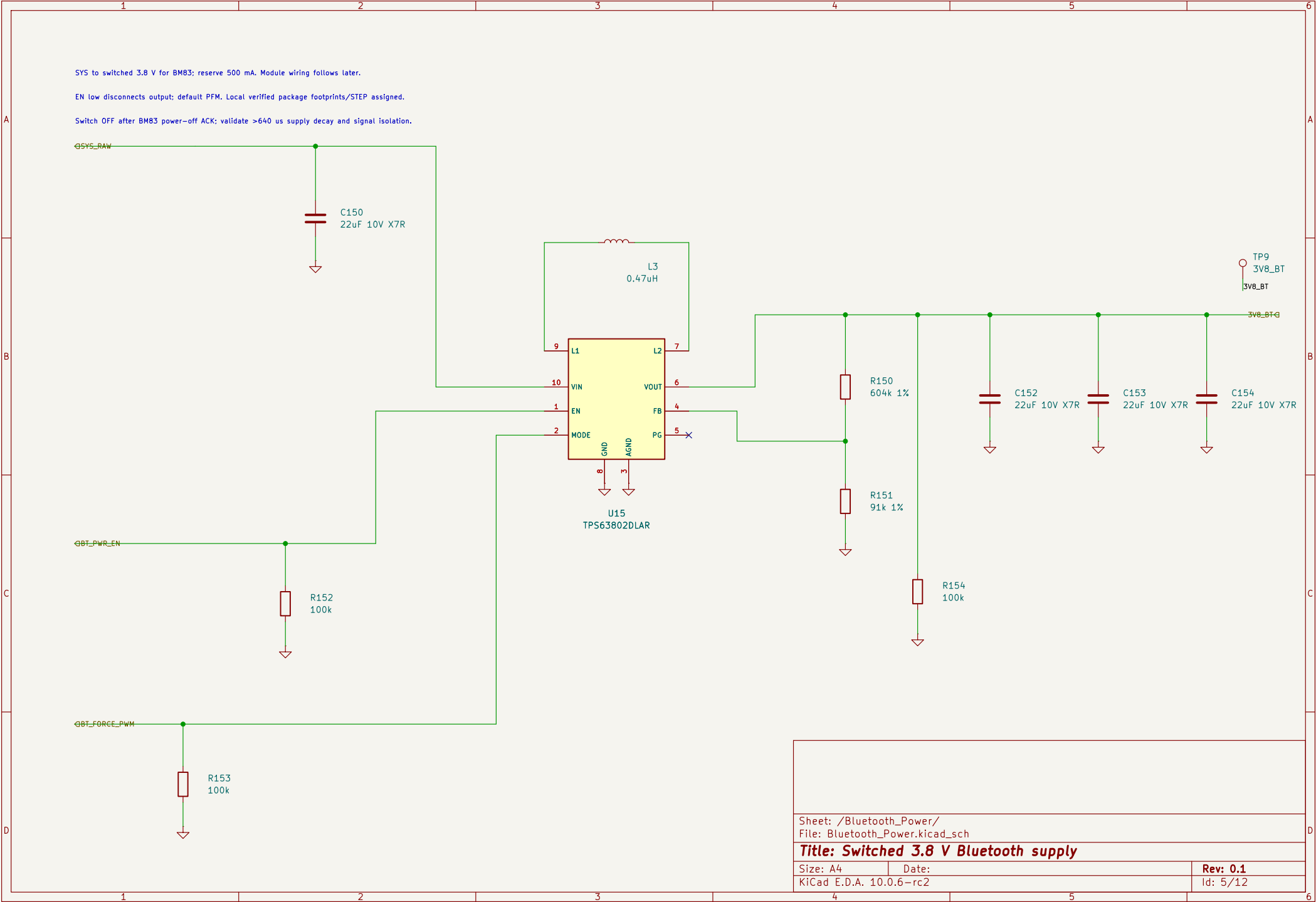
Size: A4

Date:

KiCad E.D.A. 10.0.6-rc2

Rev: 0.1

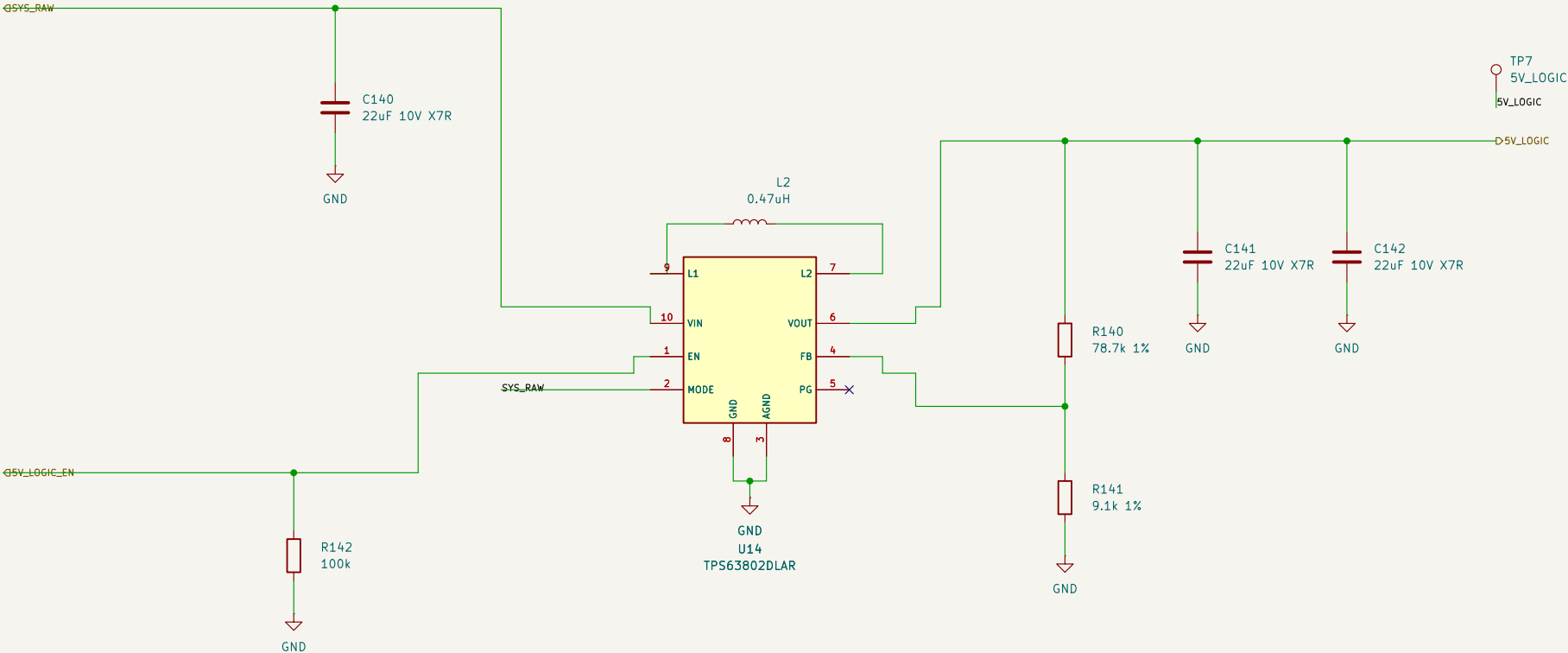
Id: 4/12



Protected 1S SYS to switched 5 V; reserve 500 mA total. Codec and other 5 V loads connect later.

EN low = true input/output disconnect; MODE tied to SYS\_RAW forces PWM after soft-start.

Steady PWM divider DC bound is within PCM2902C limits; soft-start uses PFM. Ripple and startup transient require qualification.

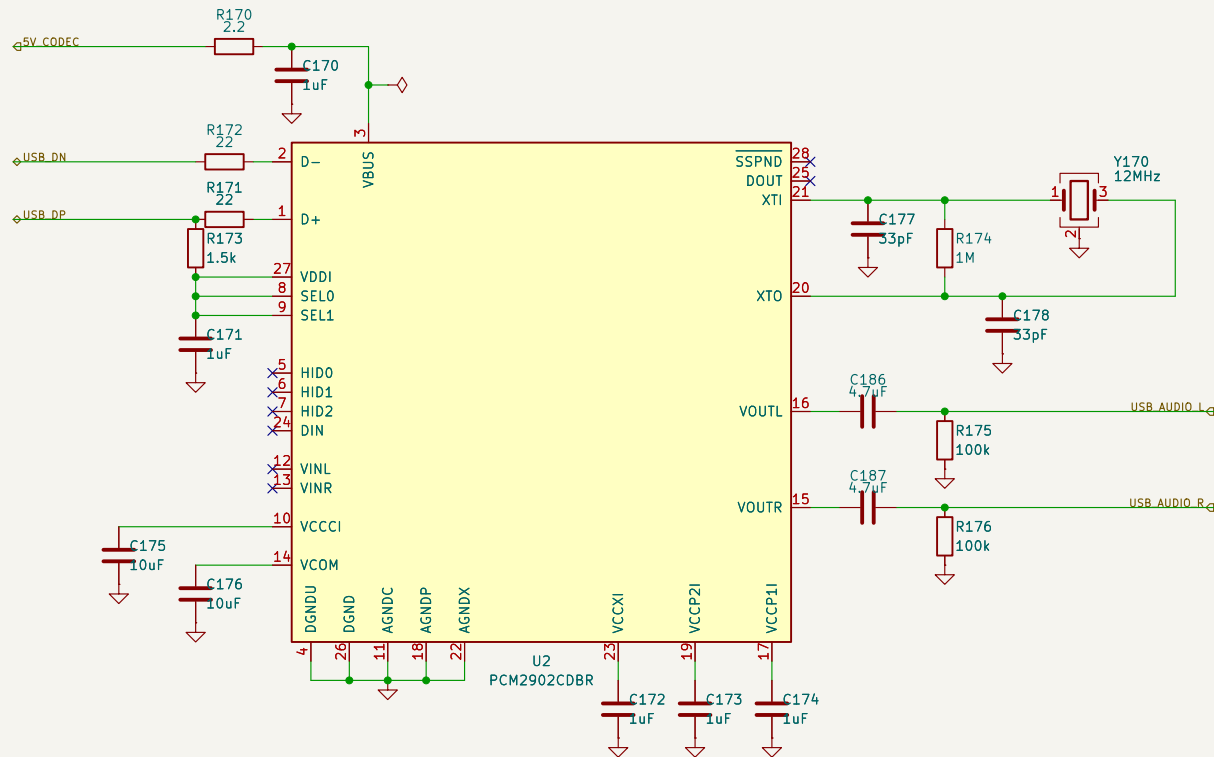




**Title: MCU and programming**

|                         |       |          |
|-------------------------|-------|----------|
| Size: A4                | Date: | Rev: 0.1 |
| KiCad E.D.A. 10.0.6-rc2 |       | Id: 7/12 |

Supply: 5V\_CODEC (U23 load switch from 5V\_LOGIC, 4.60–5.06 V worst case) feeds VBUS through 2.2R/1uF. DS range 4.35–5.25 V, 56 mA typ / 67 mA max, 250 uA suspend.  
 Codec is powered only when CODEC\_PWR\_EN (U23) and 5V\_LOGIC are on: D+ pull-up (R173 to VDDI) = USB attach. Do not enable CODEC\_PWR\_EN on DCP/unknown sources.  
 SEL0/SEL1 = high (VDDI). HID0–2, DIN, VINL/R unused. SSPND (suspend flag, active low) reserved for a later MCU input.  
 USB\_DP/USB\_DN come from J1 via D1 (root). Later source–detect mux goes between those root nets and these ports.



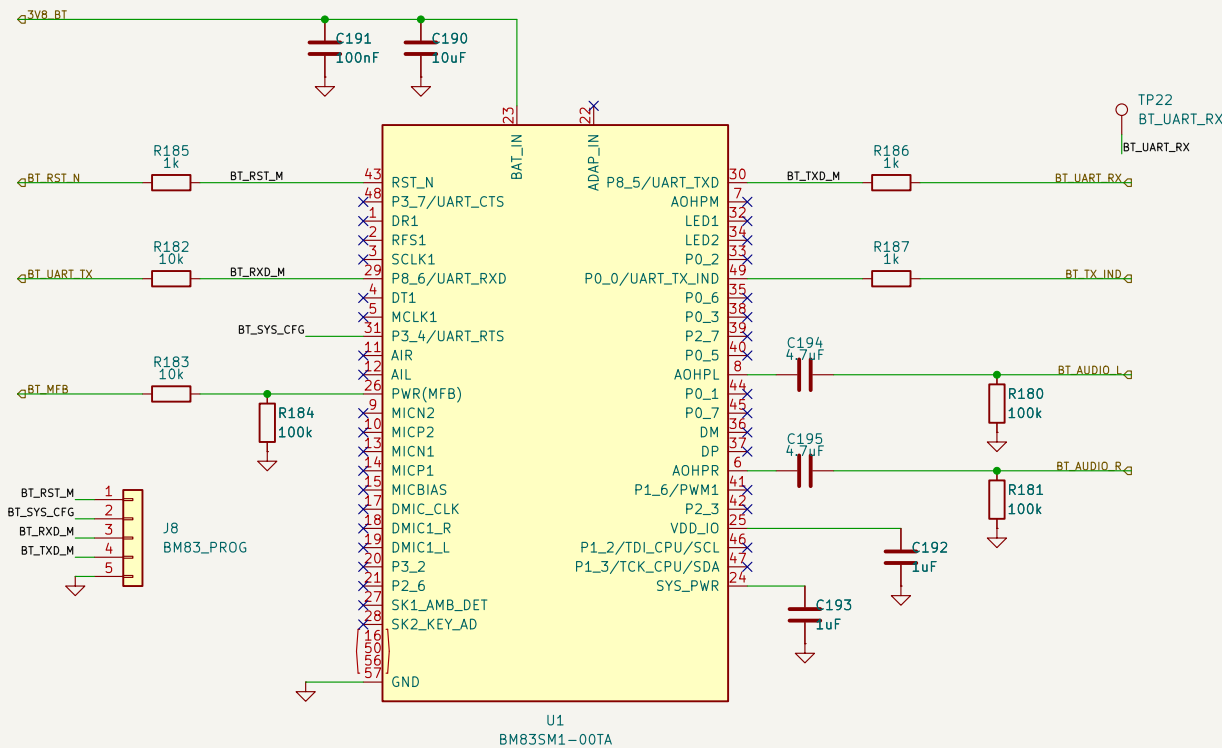
Analogue out: 0.6 VCCCI Vpp (about 2.0 Vpp, 0.7 Vrms) centred on 1.65 V; AC-coupled (HPF -0.4 Hz at 100k), >=10k load.

PCM2902C bus-powered codec from 5V\_LOGIC; USB D+/D- path; analogue out to USB\_AUDIO\_L/R

Sheet: /USB\_Audio/  
 File: USB\_Audio.kicad\_sch

**Title: USB audio**

|                         |       |          |
|-------------------------|-------|----------|
| Size: A4                | Date: | Rev: 0.1 |
| KiCad E.D.A. 10.0.6-rc2 |       | Id: 8/12 |



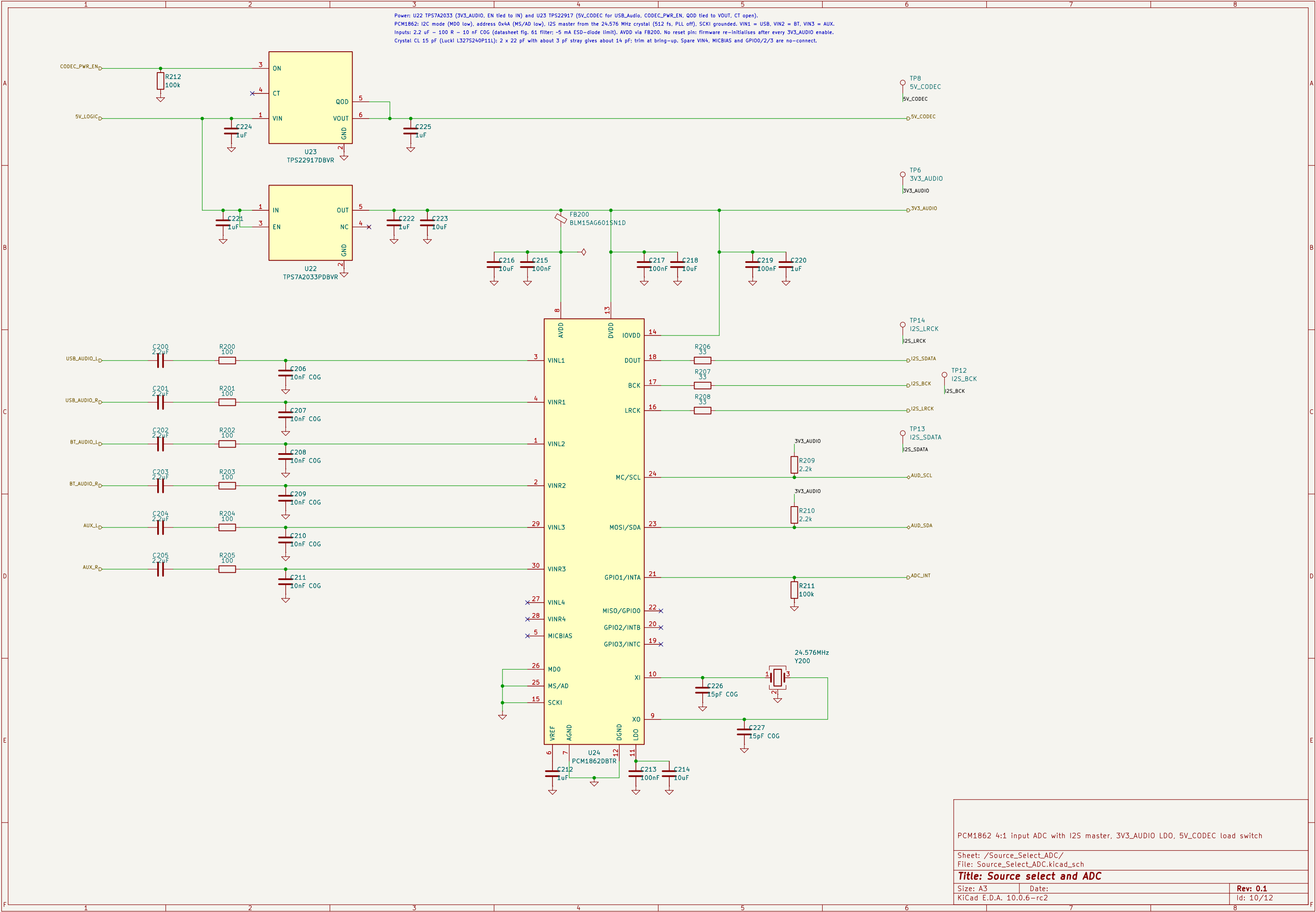
BM83SM1-00TA (DS70005402D). BAT\_IN 3.2-4.2 V from 3V8\_BT (U15); SYS\_PWR/VDD\_IO are outputs (decoupling only, do not load or feed). ADAP\_IN unused (Internal charger/USB DFU not used).  
Host mode UART from the MCU: PB\_6 RXD / PB\_5 TXD, P0\_0 = UART\_TX\_IND (wakes the MCU). MFB (PWR) = power-on/wake key. RST\_N = reset (active low, internal pull-up). Config Tool must set Host mode, P0\_0 as TX\_IND.  
Back-power: MCU lines pass 10k/1k series resistors; firmware must drive PA2/PB3 low (PB4 open-drain/hi-Z) and disable BT\_PWR\_EN only after the module power-off ACK. Module outputs are 3.0-3.6 V into 5 V tolerant MCU pins.  
Audio: single-ended DAC out (AOHPL/AOHPR) AC-coupled, 4.7 uF + 100k bleed, 0.74 Vrms full scale; AOHPM (capless sense) unused. Ports feed the later source-select sheet.  
Antenna: integrated PCB antenna, no external RF parts. Keep the antenna end of the module off the board (no copper/components/keep-out) and metal 15 mm clear; see ai-files/reports/bluetooth-notes.md.  
J8 programming/test (isUpdate/Config Tool): 1 RST\_N, 2 P3\_4 SYS\_CFG (low at reset = test mode), 3 UART\_RXD (adapter TX), 4 UART\_TXD (adapter RX), 5 GND. 3V8\_BT must be enabled first.

BM83 module: host UART, MFB/RST\_N, analogue out

Sheet: /Bluetooth/  
File: Bluetooth.kicad\_sch

**Title: Bluetooth audio**

|                         |       |          |
|-------------------------|-------|----------|
| Size: A3                | Date: | Rev: 0.1 |
| KiCad E.D.A. 10.0.6-rc2 |       | Id: 9/12 |

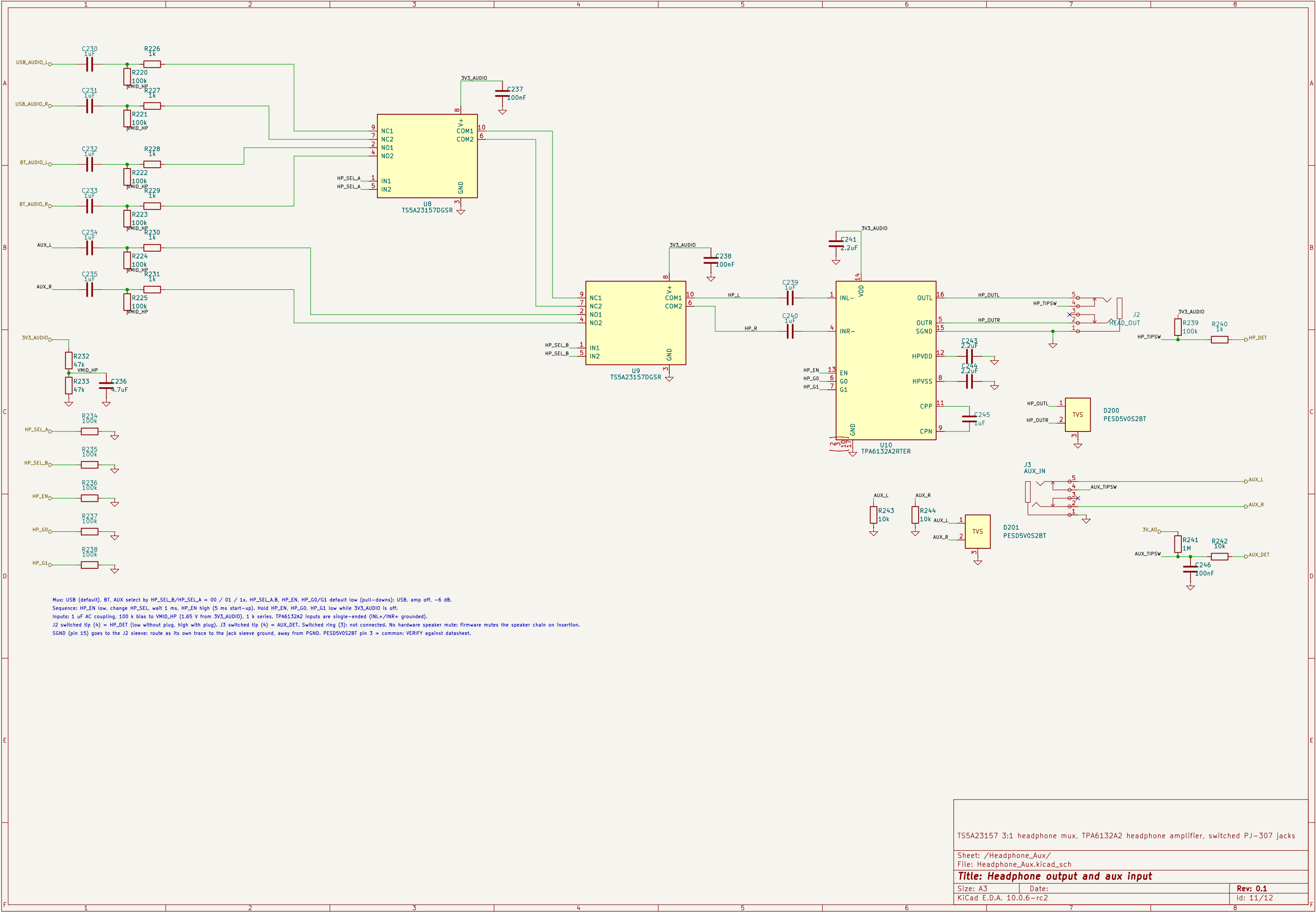


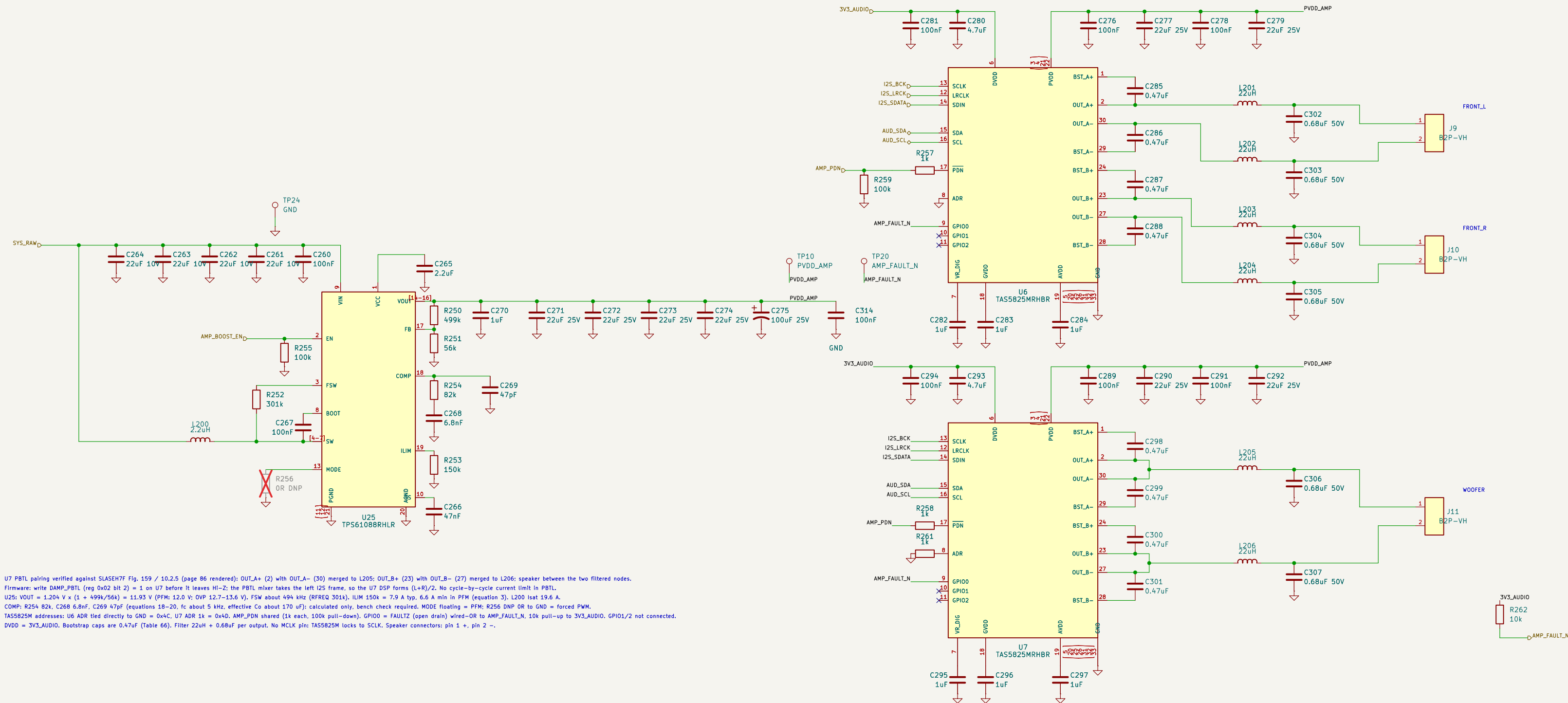
PCM1862 4:1 input ADC with I2S master, 3V3\_AUDIO LDO, 5V\_CODEC load switch

Sheet: /Source\_Select\_ADC/  
File: Source\_Select\_ADC.kicad\_sch

**Title: Source select and ADC**

|                         |       |           |
|-------------------------|-------|-----------|
| Size: A3                | Date: | Rev: 0.1  |
| KiCad E.D.A. 10.0.6–rc2 |       | Id: 10/12 |





U7 PBTl pairing verified against SLASEH7F Fig. 159 / 10.2.5 (page 86 rendered): OUT\_A+ (2) with OUT\_A- (30) merged to L205; OUT\_B+ (23) with OUT\_B- (27) merged to L206; speaker between the two filtered nodes.  
Firmware: write DAMP\_PBTl (reg 0x02 bit 2) = 1 on U7 before it leaves HI-Z; the PBTl mixer takes the left I2S frame, so the U7 DSP forms (L+R)/2. No cycle-by-cycle current limit in PBTl.  
U25: VOUT = 1.204 V × (1 + 499k/50k) = 11.93 V (PFM: 12.0 V; OVP 12.7-13.6 V); FSW about 49k kHz (RFREQ 301k); ILIM 150k = 7.9 A typ. 6.6 A min in PFM (equation 3); L200 isat 19.6 A.  
COMP: R254 82k, C268 6.8nF, C269 47pF (equations 18-20, fC about 5 kHz; effective Co about 170 uF); calculated only; bench check required; MODE floating = PFM; R256 DNP OR to GND = forced PWM.  
TAS5825M addresses: U6 ADR tied directly to GND = 0x4C; U7 ADR 1k = 0x4D; AMP\_PDN shared (1k each, 150k pull-down); GPIO0 = FAULTZ (open drain) wired-OR to AMP\_FAULT\_N; 10k pull-up to 3V3\_AUDIO; GPIO1/2 not connected.  
DVDD = 3V3\_AUDIO. Bootstrap caps are 0.47µF (Table 66). Filter 22µH + 0.68µF per output. No MCLK pin; TAS5825M locks to SCLK. Speaker connectors: pin 1 +, pin 2 -.

TPS61088 PVDD boost, two TAS5825M (stereo BTL and mono PBTl), LC output filters, speaker connectors

|                                                   |       |           |
|---------------------------------------------------|-------|-----------|
| Sheet: /Amplifiers/<br>File: Amplifiers.kicad_sch |       |           |
| Title: Amplifiers                                 |       |           |
| Size: A2                                          | Date: | Rev: 0.1  |
| KiCad E.D.A. 10.0.6-rc2                           |       | Id: 12/12 |